

Reliable Retrieval for Research Copilots

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Abstract

We study how a research copilot can retrieve papers reliably and cite only evidence that exists. This fixture text is original, written for this repository, and released under CC0 so the parser suite can run offline.

1 Introduction

Research assistants must cite evidence that exists. We describe a small pipeline that parses documents into sections and splits sections into chunks that never cross a heading boundary.

2 Method

The parser splits each page into lines, detects headings, and assigns every line to the most recent heading. Page spans are recorded from the pages a section actually touches. Unknown pages remain null rather than invented.

Table 1: Chunk policy

target 450 cap 600 overlap 60

policy fixed-window version v1

3 Results

The fixture parses into an abstract, three numbered sections, one table and one reference list. Rerunning the parser on identical bytes reproduces the same sections, checksums and chunk identifiers.

References

[1] A. Lovelace. Notes on the Analytical Engine. 1843.

[2] G. Hopper. The Education of a Computer. 1952.